What can we learn from over 200 million protein structures?

Characterization of rare and novel AlphaFold structural space

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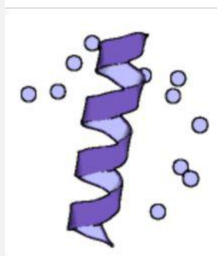
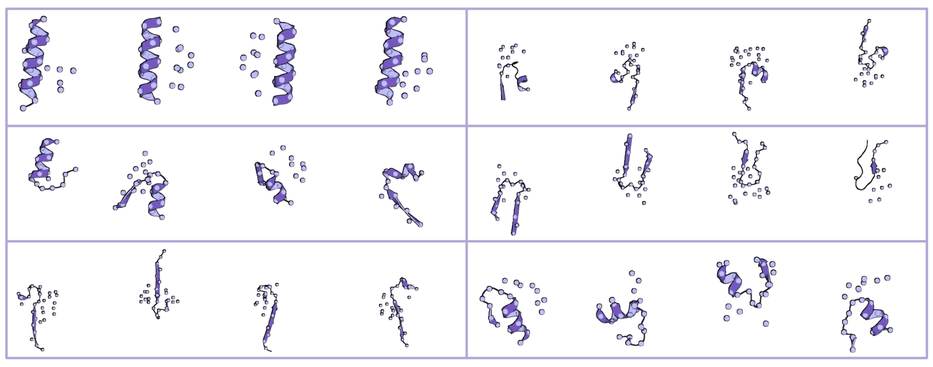
- Fragment protein structures into overlapping k-mers
- Compare 3D fragments using **rotation-invariant moments**
- Discretize moments into 1024 **"shape-mers"**
- Shape-mers ~ k-mers ~ **words in a document**

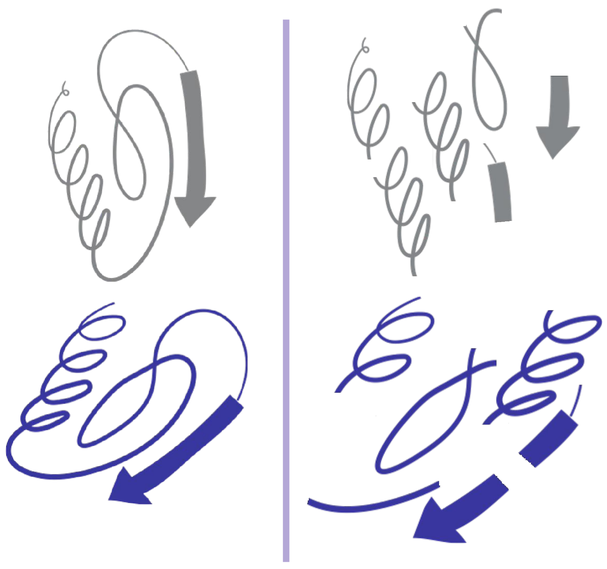




Geometricus

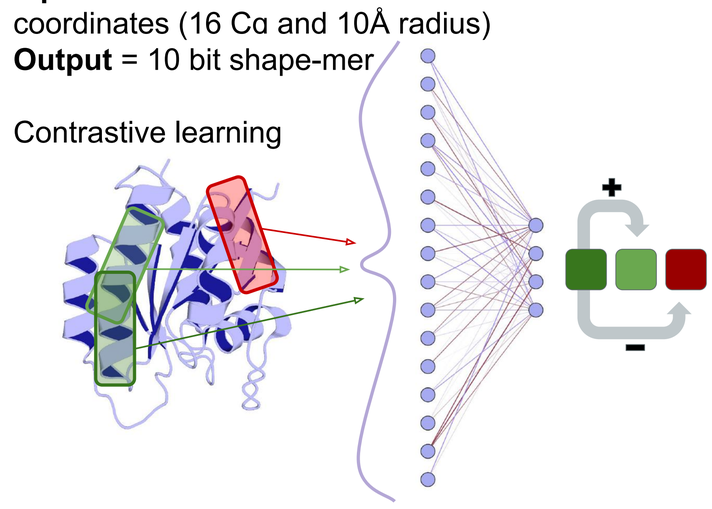
Rotation-invariant moments

50 milliseconds / protein

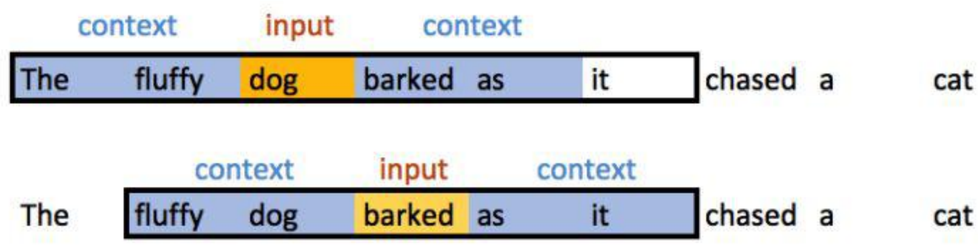


Input = Rotation-invariant moments of a set of coordinates (16 Cα and 10Å radius)
Output = 10 bit shape-mer

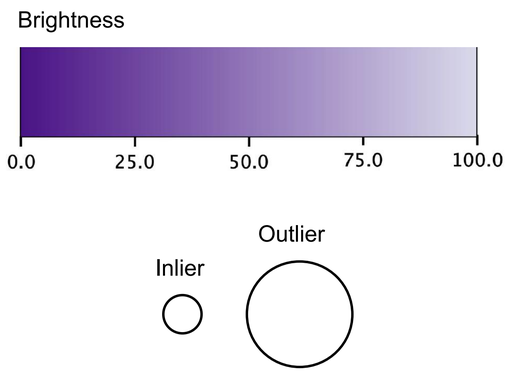
Contrastive learning



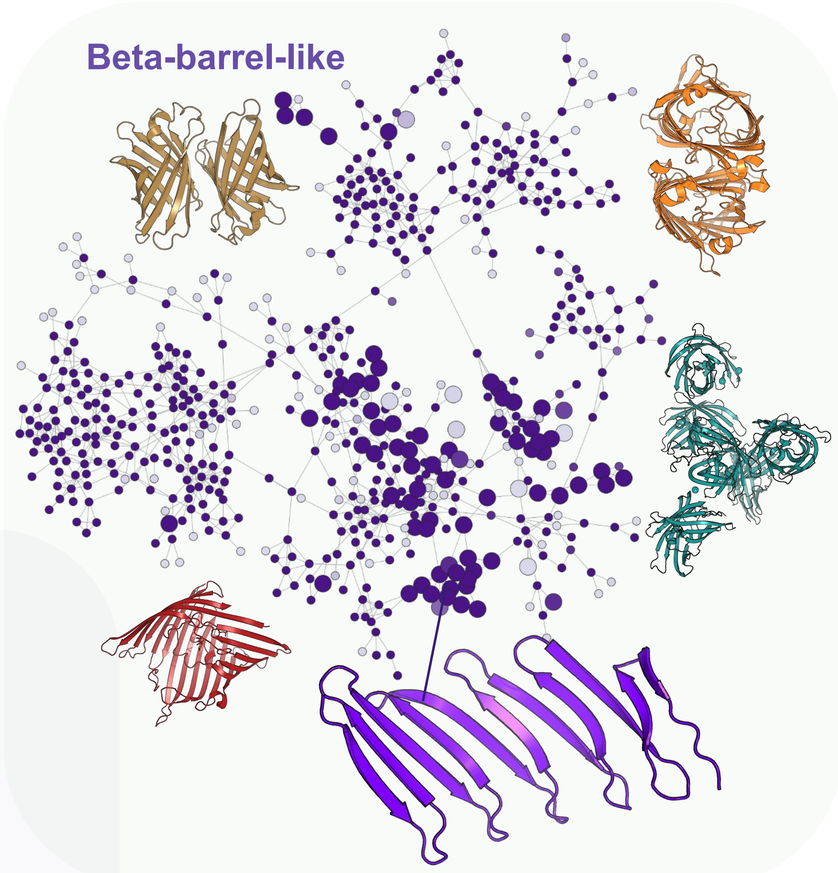
- **Natural Language Processing** approaches
- **Word2Vec** - embed word with context
  - 1024 length embedding per shape-mer
  - 1024 length embedding per protein
- **Outlier** detection - compare to PDB
- Analyse **200 million protein structures** in days
- 1M UniRef50 representatives with >95 pIDDT



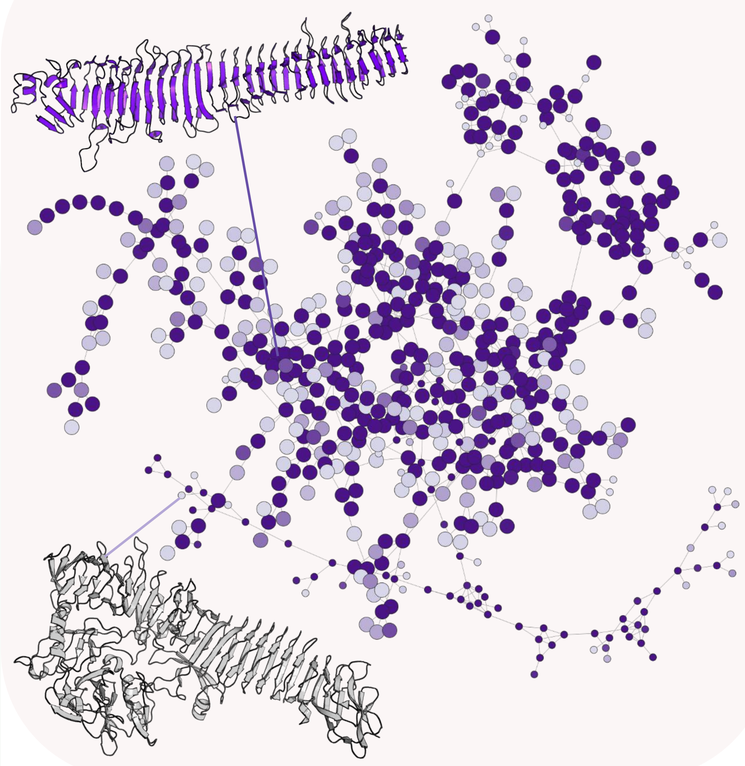
- Look for proteins that are **dark** - unannotated, function unknown
- Connect dark proteins to proteins with **similar** structure embeddings
- Look for **structural outliers** compared to the PDB



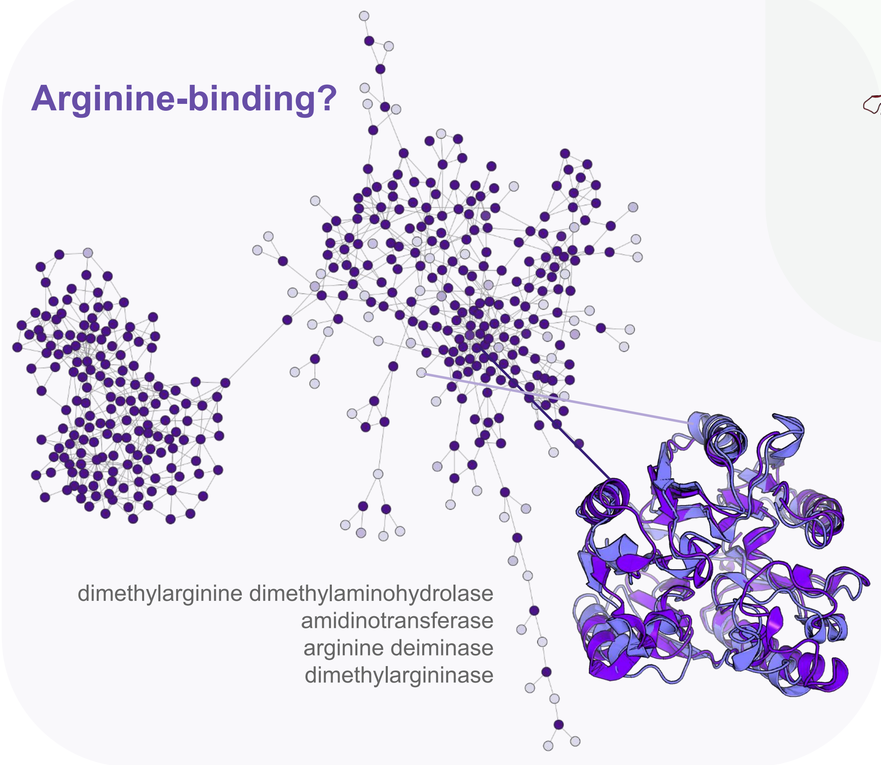
Beta-barrel-like



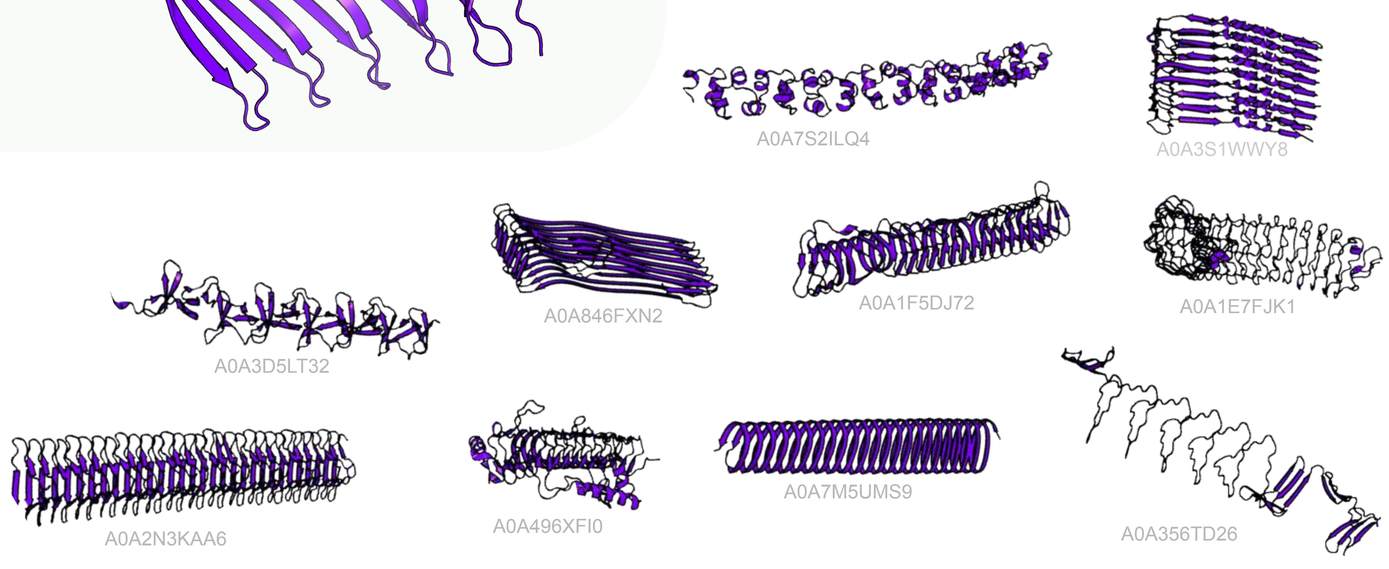
Adhesins?



Arginine-binding?



Structural outliers



What's next?

Fast **search** and alignment ||| Structural **motif** definition ||| **Language model** trained on millions of structures

References
Durairaj, Janani, et al. "Geometricus represents protein structures as shape-mers derived from moment invariants." *Bioinformatics* 36.Supplement_2 (2020): i718-i725.
Jumper, John, et al. "Highly accurate protein structure prediction with AlphaFold." *Nature* 596.7873 (2021): 583-589.
Varadi, Mihaly, et al. "AlphaFold Protein Structure Database: Massively expanding the structural coverage of protein-sequence space with high-accuracy models." *Nucleic acids research* 50.D1 (2022): D439-D444.
Akdel, Mehmet, et al. "A structural biology community assessment of AlphaFold 2 applications." *BioRxiv* (2021).

See Also:

Joana Pereira et al. "What is hidden in the darkness? A large-scale approach to make sense of all natural unknown proteins"